Christof Teuscher

ECE 410/510: Hardware for AI and ML, Spring 2026

Week 2: Recap, outlook, and reminders

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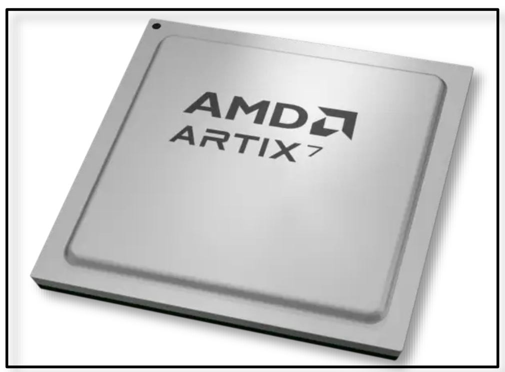
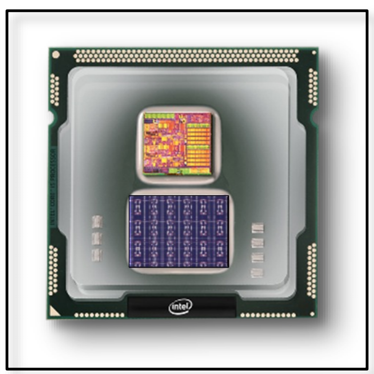
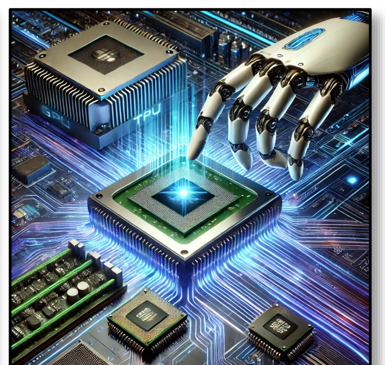
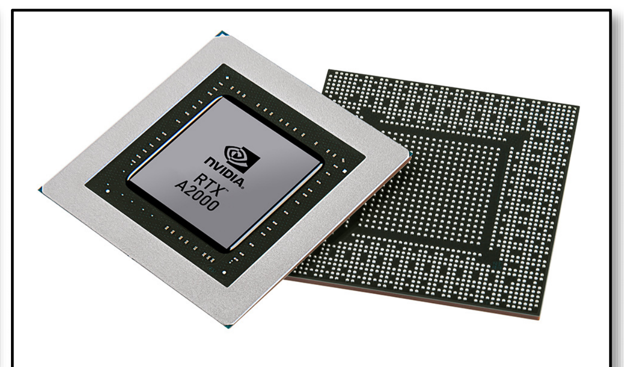


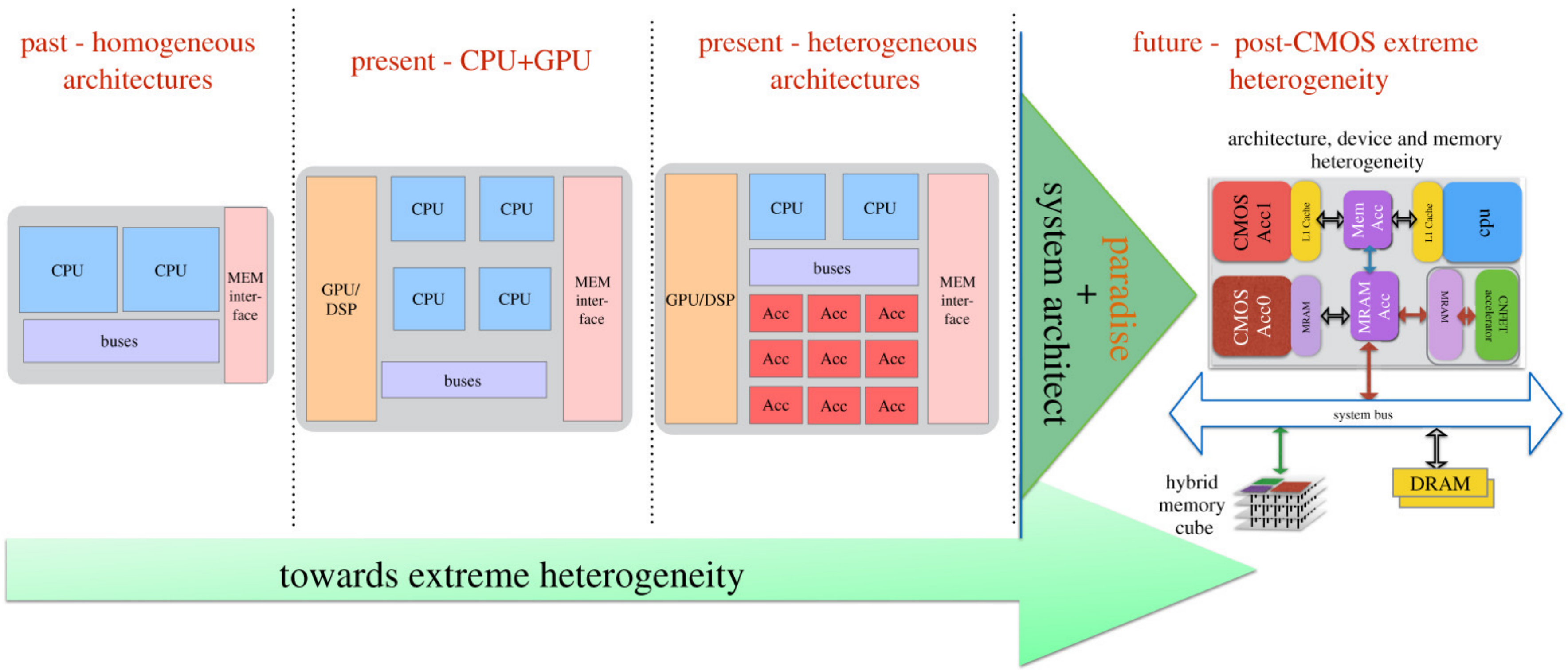






What did you learn last week?

The path to extreme heterogeneity



https://royalsocietypublishing.org/doi/10.1098/rsta.2019.0061

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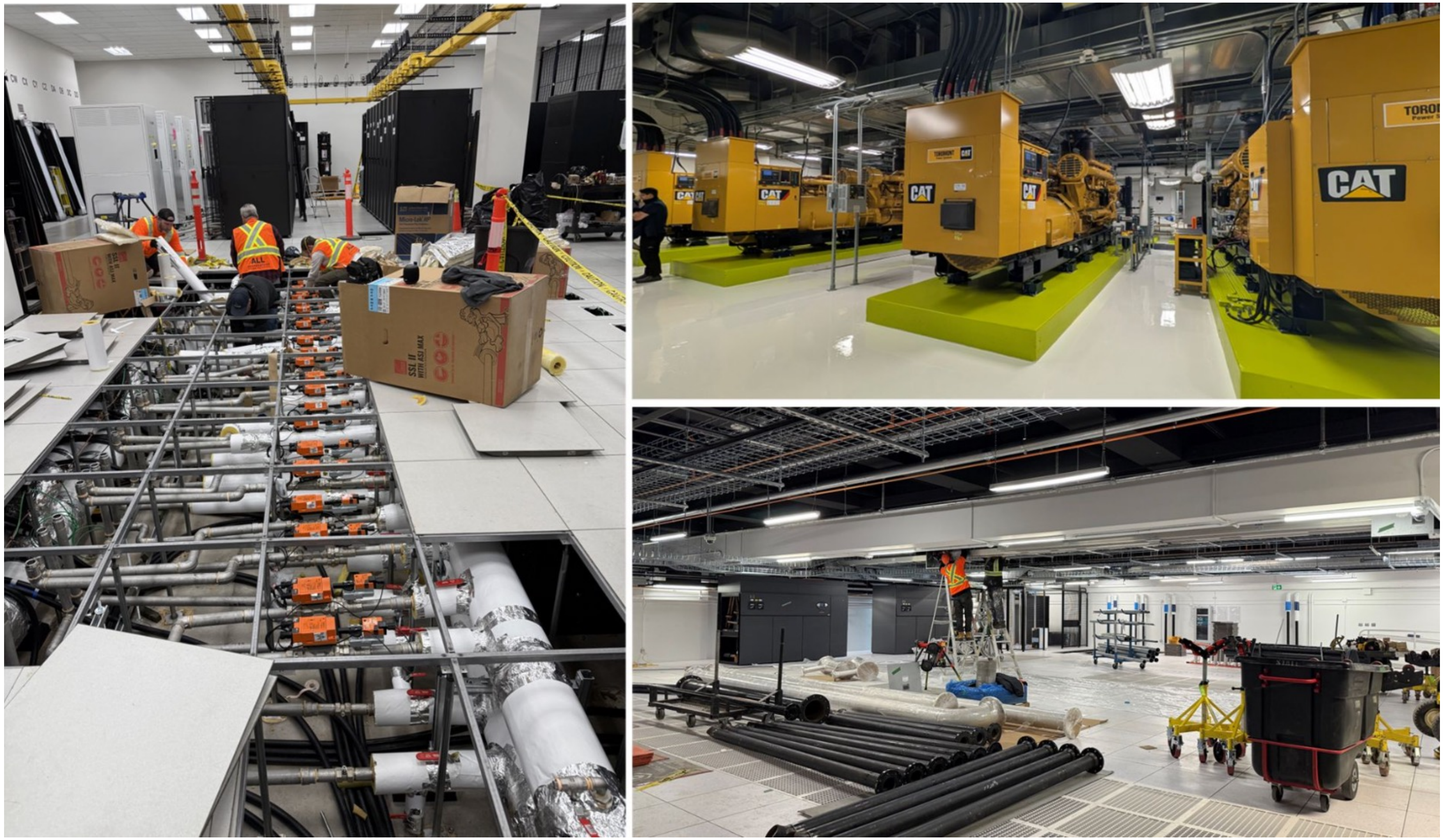
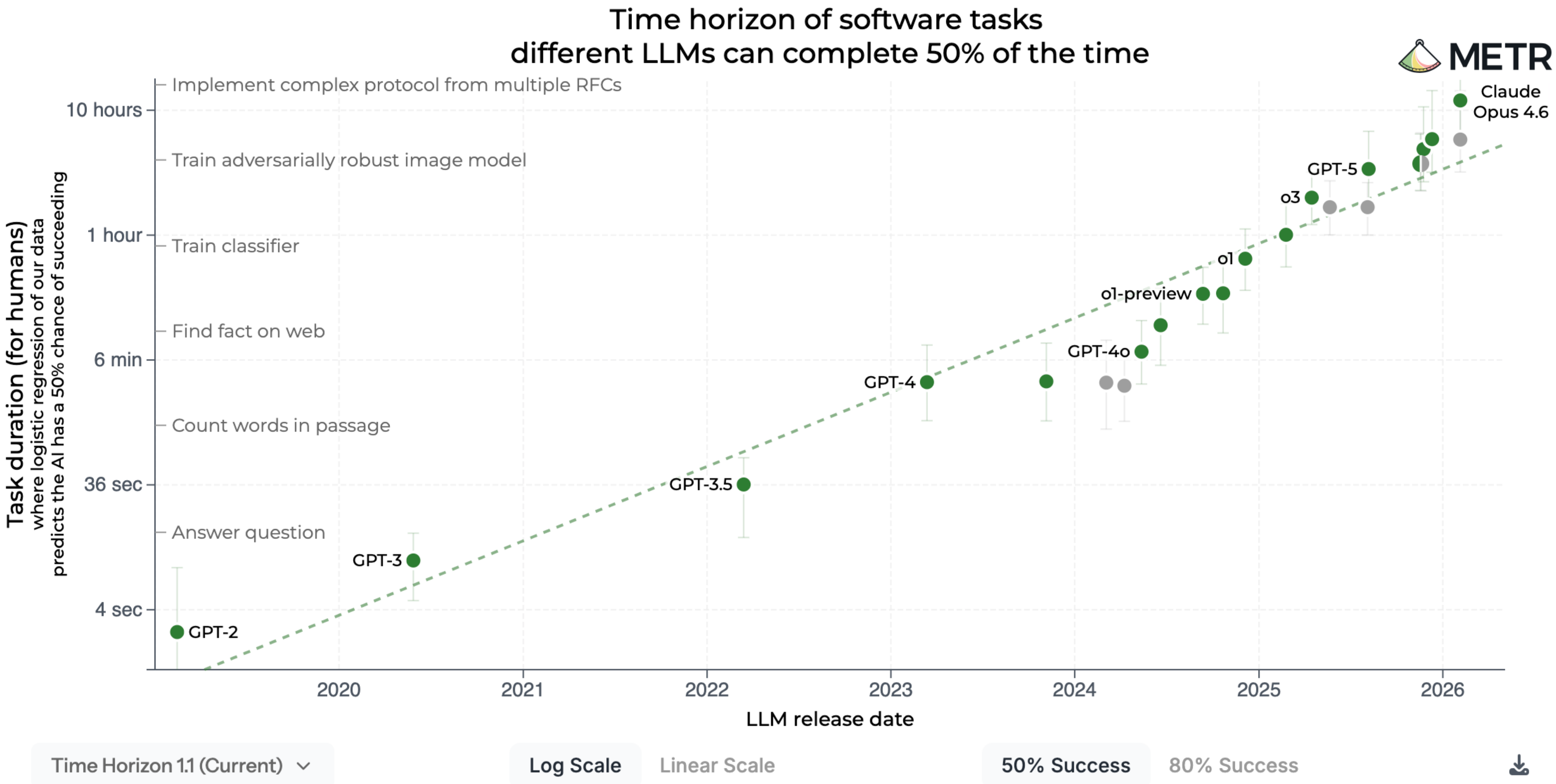


Cerebras datacenter



Over the past 6 years, the length of a task an AI can do has doubled every 7 months.

https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks

HLE is generally seen as the toughest test of reasoning and knowledge that can be given to LLMs.

It's 2500 questions over multiple disciplines, answers are largely offline, and many are kept entirely secret so they can't be talked about online.

They are set by the world's top experts in the respective subjects, with prizes given to those who set questions AI cannot answer.

Just over a year ago, the top frontier model, GPT-4o, achieved about 3%. Most recently, Gemini 3.1 achieved about 45%.

Why does this matter? Look at the name of the benchmark. There is no tougher exam that could be given for humans to complete.

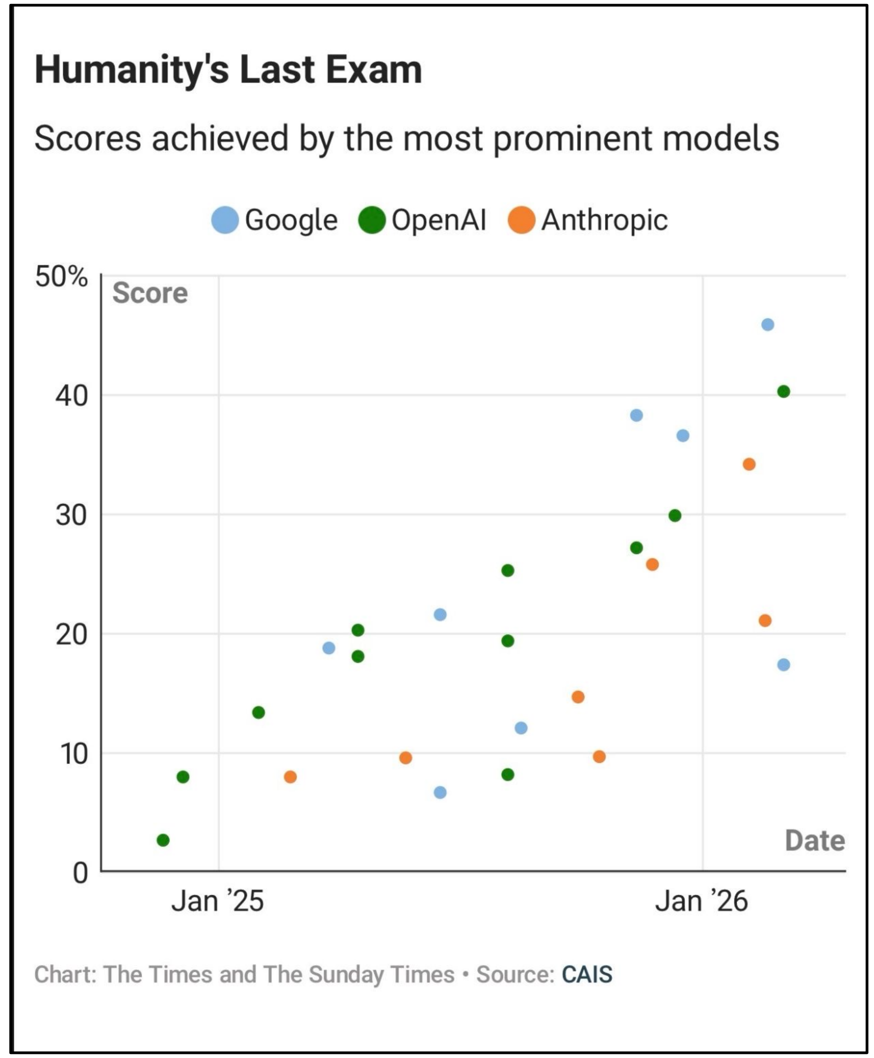
Which means that, for the testing of reasoning and knowledge ability (aka what a paper-based education is for), AI is already far in advance of the most brilliant humans (even the world's top experts cannot achieve more than 10%, such are the deep specialisms required).

And soon it will have no more human-answerable questions to answer.

And yet we perpetuate this in schools:

'Now, class, turn to page 43 and do questions 1-10.'

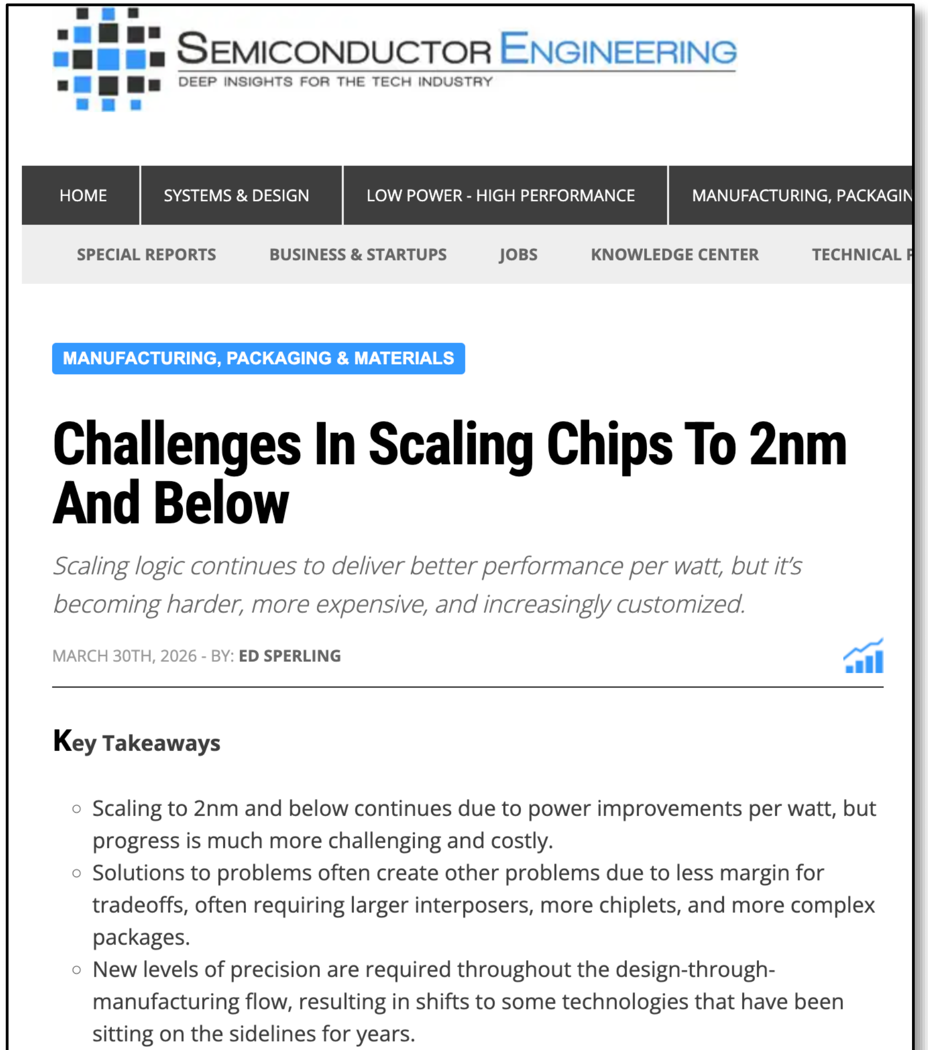
Something's gotta give.



https://www.linkedin.com/posts/darren-coxon_researchers-believe-were-only-one-year-away-share-7444233040941379584-7cJ1

2nm and below…

PPAC still rules!



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MANUFACTURING, PACKAGING & MATERIALS

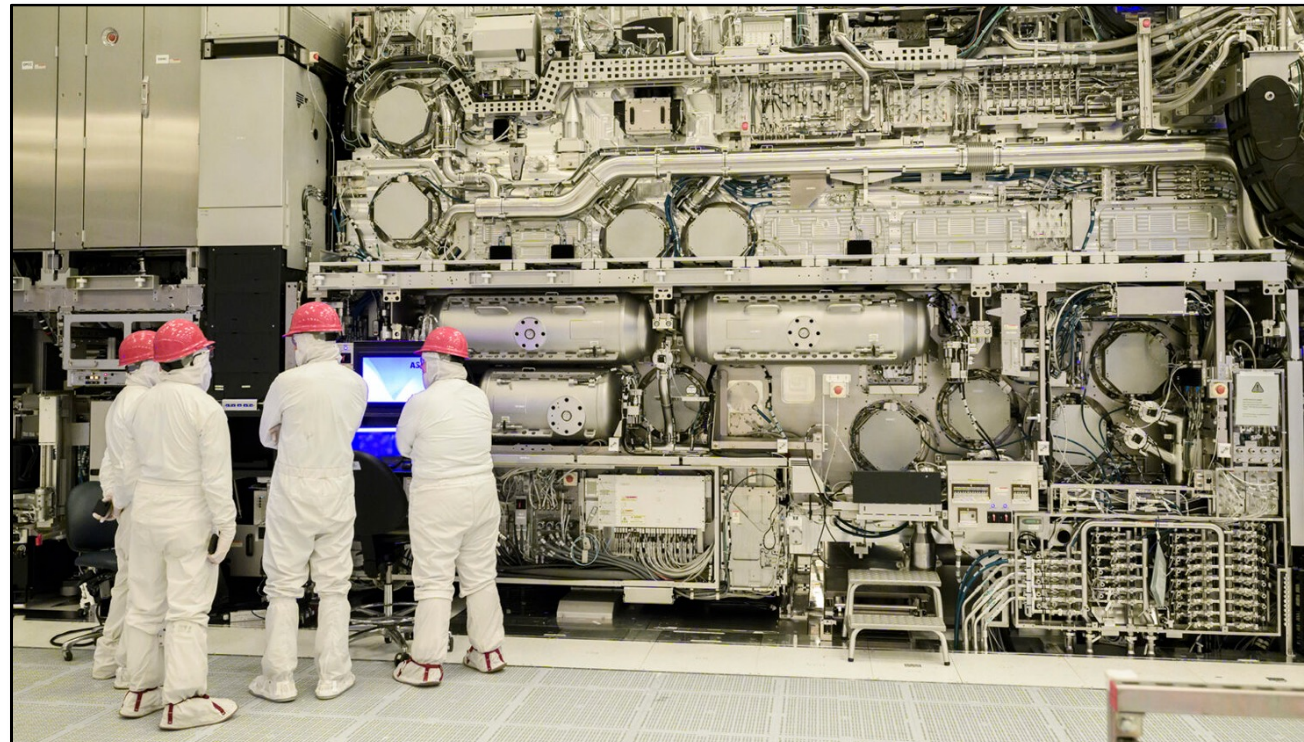
Challenges In Scaling Chips To 2nm And Below

Scaling logic continues to deliver better performance per watt, but it's becoming harder, more expensive, and increasingly customized.

MARCH 30TH, 2026 - BY: ED SPERLING

Key Takeaways

- Scaling to 2nm and below continues due to power improvements per watt, but progress is much more challenging and costly.
- Solutions to problems often create other problems due to less margin for tradeoffs, often requiring larger interposers, more chiplets, and more complex packages.
- New levels of precision are required throughout the design-through-manufacturing flow, resulting in shifts to some technologies that have been sitting on the sidelines for years.



Intel's 165-ton high-NA EUV scanner, priced at $350+ million.
Image source: Intel Foundry

https://semiengineering.com/challenges-in-scaling-chips-to-2nm-and-below

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Future chip designers

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SYSTEMS & DESIGN

Can A Computer Science Student Be Taught To Design Hardware?

To fill the talent gap, CS majors could be taught to design hardware, and the EE curriculum could be adapted or even shortened.

FEBRUARY 17TH, 2026 - BY: **LIZ ALLAN**

https://semiengineering.com/can-a-computer-science-student-be-taught-to-design-hardware

Key Takeaways

- New approaches are being devised and tested to address the talent shortage.
- Leveraging AI in design tools will help engineers become more efficient, and potentially could reduce the time it takes to train engineering students.
- EDA companies are looking at whether it's possible to train computer science and software engineers to become hardware engineers.

How to Enable SW Programmers to Do Chip Design?

- Start with the languages they are using
  - Leverage high-level synthesis
- Automated code transformation and optimization to RTL
  - With deep learning
  - With polyhedral optimization and transformation ...
- An agentic approach to orchestrate various design and optimization steps
- Create an "App Store" (see Horowitz's keynote)

Engineering mindset

- **Make things better!** This is the essence of computer engineering: make things better, faster, more efficient, etc.

- See what your LLM generates as a starting point.
- **Make it better**. Use the LLM to make it better.
- **Make it perfect!**

- Use the LLM for explanations: “Explain me what happens on lines 3-10 and why”
- Use your LLM as a thinking tool.

Codefest #2 preparation

- Finalize your AI/ML algorithm/workload you want to pick.
- It should be a non-conventional workload that will benefit from specialized hardware.
- Use Google Scholar to find relevant literature on current state-of-the-art
- You will be designing your own accelerator chiplet for that workload.